gbsv Mini-Challenge 1

2026-02-24

General Information

In this mini-challenge, a profile of a country (Steckbrief) will be compiled in the form of experiments in foundations in image and signal processing to assess aspects of LO1-LO3. Each student will submit a unique solution with individual data and analysis. This means you will choose images and signals themed suitably for your selected country and the tasks assigned to you. Unless otherwise mentioned, existing libraries may be used.

Note: Certain tasks are kept open-ended. Set your own frame for the limitation or ask if the boundaries are not clear to you. As a rule of thumb one day should take you around 2h +/-30min per day. Appropriate data selection, pinpoint creativity, moderate diversity, and critical thinking are sought rather than extensive submissions.

Version Individual Rhythm

- Conduct your MC according to your own schedule/rhythm
- Mandatory check-in (review meeting): MC is only graded if a check-in was conducted.

Version 15-Day Challenge

- Profit from regular feedback
- Earn 1 bonus credit per submitted day if you submit at least 10 days
- Bonus credits: up to +0.33 added to final grade (scaled to 15; capped at 6.0 for maximum grade)
- Revisions are allowed until day 15 if changes are clearly highlighted in color. Code does not need to be colored, but a colored text tag should mark start and end of code changes.
- Submission deadline: all entries must be submitted by day 15 (per challenge)

Submission and Grading

- Submission are conducted via Microsoft Teams Assignments in the gbsv Teams.
- Deliverables:
    - Submit HTML export of your submission (one notebook per section 1.1-1-3)
    - Link to code and results incl. signals/images, if applicable
    - No zip files are accepted!
- Make sure that audios are playable within notebook exports
- See grading rubrics, gbsv interpretations and indicators for strong vs. weak performance
- Submissions received after the deadline or submitted via email will not be considered.
- Assessment is based on the quality of your technical framing and justification, not on whether you adopt a provided example use case or develop your own.

1.1 Sampling Theorem

Learning Objectives

- Conceptual: Apply the Nyquist-Shannon theorem to explain aliasing in a domain-specific context.
- Methodological: Identify a domain-relevant signal where sampling affects information fidelity and derive its required sampling rate from the bandwidth.
- Technical: Implement and compare sampling above, at, and below the Nyquist rate.
- Analytical: Quantitatively assess sampling distortion using appropriate metrics.
- Reflective: Evaluate practical risks and real-world implications of incorrect sampling decisions.

Day 1 – Data & Domain

On this day, you will define the conceptual and application framework of this MC task. First, describe a realistic use case in which sampling plays a critical role and briefly justify why it is relevant to your selected country. Then formulate a precise problem statement that explicitly addresses the sampling rate, identifies what may go wrong if it is chosen inadequately, and clarifies which aspect of signal fidelity is essential in your context. Next, state the experimental objective by defining what you intend to investigate regarding sampling behavior and explain how this investigation helps address the technical vulnerability identified in your problem statement.

Then select and describe a representative one-dimensional signal suitable for your use case, including its physical origin, unit of measurement, acquisition method, and expected frequency characteristics.

Interpret the Nyquist frequency and Nyquist rate for your specific signal and application. Base your explanation on the physical meaning and the observable temporal or spatial variations in your data. Discuss practical consequences of too low and unnecessarily high sampling rates in your use case. No mathematical derivations or implementations are required at this stage.

Start a new notebook for this block.

Reference examples: The following use cases are provided for orientation. You may select one of them and adapt it to your chosen country, or define a suitable alternative within your selected domain.

Ocean Wave Monitoring

Use Case: At a coastal holiday destination, wave height is continuously recorded to assess swimming safety and characterize prevailing sea conditions. The ocean surface motion consists of periodic swell components superimposed with shorter wind-driven waves.

Problem Statement: The wave height signal contains multiple frequency components that describe the energy and dynamics of the sea state. If the sampling rate is chosen too low, faster wave components may be aliased and appear as misleading lower-frequency oscillations, potentially distorting the interpretation of wave conditions. Conversely, excessively high sampling increases storage and transmission demands without adding meaningful information beyond the physical bandwidth of the signal. The critical requirement is the faithful digital representation of the relevant wave frequency range without distortion.

Audio Engineering – Speech Recording

Use Case: Digital speech recordings are used in communication and voice-processing systems. Speech signals contain broadband frequency components that determine intelligibility and phonetic structure.

Problem Statement: Speech signals include frequency components that are essential for preserving intelligibility and phoneme discrimination. If the sampling rate falls below the required threshold, higher-frequency components may be aliased or distorted, altering spectral characteristics and reducing clarity. At the same time, unnecessarily high sampling increases data volume and computational load without improving perceptual quality. The central issue is ensuring accurate spectral representation of the relevant speech bandwidth in the digital domain.

Fitness Tracker During Hiking Holidays

Use Case: During a hiking holiday, a wearable fitness tracker records acceleration signals and optical pulse measurements to monitor movement and cardiovascular activity. These signals contain rhythmic physiological and biomechanical components.

Problem Statement: Acceleration and pulse signals exhibit dominant periodic components, higher-order harmonics, and short-term variations caused by changes in activity. If the sampling rate is insufficient, rapid physiological fluctuations or harmonic components may be aliased or attenuated, leading to inaccurate representation of cadence or heart rate dynamics. Excessively high sampling, however, increases energy consumption and reduces battery life in wearable devices. The key requirement is the reliable digital capture of relevant physiological frequency content within practical device constraints.

Wind Turbine Vibration Monitoring

Use Case: Vibration sensors installed in a wind turbine record mechanical oscillations associated with rotor rotation and structural dynamics. These measurements are used to characterize operational behavior.

Problem Statement: Mechanical vibration signals contain characteristic frequency components related to rotational speed and structural resonance modes. If the sampling rate is too low, higher-frequency vibration components may be aliased and misrepresented, distorting spectral interpretation of turbine dynamics. If the sampling rate is excessively high, data storage and processing requirements increase without capturing additional physically meaningful information. The essential requirement is an accurate digital representation of the vibration bandwidth relevant to mechanical behavior.

Day 2 – Methodological

On this day, you will establish the theoretical and methodological foundation for your sampling analysis. Formally state the Nyquist–Shannon Sampling Theorem and explain its underlying assumptions, particularly the requirement of band limitation.

Based on the signal defined in Day 1, determine its bandwidth and clearly justify how this bandwidth is identified (e.g., analytical derivation, known signal properties, or justified approximation). From this bandwidth, explicitly derive both the Nyquist frequency and the corresponding Nyquist rate. State the minimum required sampling rate mathematically and document each step of your reasoning.

In addition, explain how you will later vary the sampling rate in a structured way to investigate sampling behavior and potential aliasing effects. Justify why these parameter choices are methodologically appropriate for your signal and use case.

The focus of this day is on mathematical rigor and methodological transparency; no simulations, visualizations, or experimental demonstrations should be included.

Day 3 – Technical Implementation

On this day, you will implement the computational framework required to investigate sampling behavior in your defined signal. Based on the methodological decisions from Day 2, generate a high-resolution reference version of your signal that serves as a continuous approximation.

Implement discrete sampling at systematically chosen sampling rates, including at minimum: a rate above the Nyquist rate, exactly at the Nyquist rate, and below the Nyquist rate. Ensure that all parameters are clearly defined and traceable.

Visualize the sampled signals in the time domain and, where appropriate, in the frequency domain. Explicitly demonstrate aliasing effects in the undersampled case. The objective is to produce a technically correct implementation that reflects the methodological framework defined previously.

No quantitative evaluation or extended interpretation is required at this stage; the focus lies on clean implementation, structured experimentation, and traceability.

Day 4 – Evaluation

On this day, you will systematically evaluate the sampling configurations implemented in Day 3. Using 1-2 quantitative evaluation approaches (e.g., numerical metrics and/or quantitatively interpretable visualizations), assess the differences between the high-resolution reference signal and the sampled versions. Suitable approaches may include reconstruction error, frequency deviation, amplitude distortion, or measurable spectral characteristics.

The choice of evaluation approach must be explicitly justified. Explain why the selected metric or extracted visualization quantity is appropriate for your signal characteristics and for the objectives defined in your use case. Clearly specify what aspect of signal quality is captured (e.g., amplitude fidelity, frequency preservation, distortion).

Compare the cases above, at, and below the Nyquist rate in a structured and traceable manner using a clearly defined baseline configuration. Quantify performance differences consistently across all configurations and report values with appropriate units or normalization. Identify the sampling rate at which distortion becomes significant and quantify the magnitude of information loss.

If the signal represents audio data, playback of the sampled signals may be included as a supplementary perceptual evaluation. In this case, briefly describe audible differences (e.g., aliasing artifacts, distortion, loss of clarity) in relation to the quantitative results. Audio playback does not replace the required metric-based or quantitatively measurable evaluation but may support interpretation in use cases where perceptual quality is relevant.

The focus of this day is on objective, quantitative assessment and structured comparison. No new simulations, theoretical derivations, or interpretative discussion should be introduced.

Day 5 – Analytical & Communication

On this day, you will observe, interpret, and discuss your findings in direct relation to your initial problem statement and defined use case. Based on the quantitative evaluation from Day 4, analyze how different sampling rates affect signal fidelity and how these effects translate into domain-specific consequences.

Structure your analysis into observations, interpretation, and discussion. Discuss the practical applicability of your results: Which sampling rate is appropriate and realistic in your chosen context, and why? What risks arise if the sampling rate is chosen incorrectly? Critically assess how the theoretical assumptions of the Nyquist-Shannon theorem align with real-world conditions in your application domain.

Present your analysis in a concise, technically precise, and logically structured form (approximately 180-220 words). All arguments must clearly connect back to the original problem statement. No new simulations or derivations should be introduced.

1.2 Correlation

Learning Objectives

- Conceptual: Apply auto-correlation and cross-correlation for one-dimensional signals to explain periodicity and similarity detection in a domain-specific context.
- Methodological: Define appropriate parameters and evaluation criteria for detecting recurring patterns and signal segments.
- Technical: Implement and visualize auto-correlation and cross-correlation in a structured and traceable manner.
- Analytical: Quantitatively assess detection performance and assess cross-correlation robustness under realistic perturbations.
- Reflective: Evaluate the reliability and limitations of correlation methods in practical signal analysis.

Day 6 – Data & Domain

On this day, you will define the domain context and problem statement for your correlation analysis. Select a country-related one-dimensional signal that contains recurring structures and identifiable segments. Clearly formulate a technically meaningful use case in which detecting periodicity or identifying specific signal patterns is relevant.

Describe the origin and physical meaning of the signal, explain why recurring patterns or repeated structures are expected, and justify why correlation analysis is an appropriate method for your defined problem. The focus of this day is on structured problem framing, signal justification, and domain relevance. No implementation or quantitative analysis is required.

Start a new notebook for this block.

Reference examples: The following use cases are provided for orientation. You may select one of them and adapt it to your chosen country, or define a suitable alternative within your selected domain.

Ocean Wave Monitoring – Periodicity Detection

Use Case: At a coastal holiday destination, wave height is continuously recorded to assess swimming safety and to characterize recurring swell systems. Ocean surface motion is influenced by periodic swell components and shorter wind-driven waves.

Problem Statement: Wave height signals contain recurring oscillatory structures corresponding to dominant swell periods and wave groups. Identifying these periodic components is important for understanding sea state conditions and assessing potential hazards. Additionally, comparing signals from two spatially separated measurement points can reveal similarities in wave patterns and propagation effects. The task is to analyze recurring structures and similarities within and between wave signals using correlation methods in a physically meaningful way.

Speech Signal – Repeated Pattern Detection
Use Case: In a voice-processing application, a speech recording is analyzed to identify recurring acoustic structures such as repeated words or quasi-periodic voiced segments.

Problem Statement: Speech signals contain repeated temporal structures, including pitch periodicity in voiced sounds and recurring phonetic segments. Detecting these patterns is relevant for speech segmentation and recognition tasks. However, speech is inherently non-stationary and may contain

noise and articulation variability. The problem is to identify and interpret recurring structures and segment similarities in a speech signal using correlation-based analysis.

Fitness Tracker – Step and Heartbeat Cycles
Use Case: During a hiking holiday, a wearable fitness tracker records acceleration and optical pulse signals to monitor physical activity. These signals reflect repeated biomechanical and physiological cycles.

Problem Statement: Acceleration signals from walking or running contain recurring step cycles, while pulse measurements contain repeating heartbeat intervals. Detecting these periodic structures is essential for estimating cadence and heart rate. In real-world conditions, motion artifacts and physiological variability may alter signal shape and amplitude. The problem is to identify recurring patterns and similarities within these signals and interpret their physical meaning using correlation analysis.

Wind Turbine Vibration Monitoring – Rotational Periodicity
Use Case: Vibration sensors installed in a wind turbine record mechanical oscillations caused by rotational dynamics and structural behavior.

Problem Statement: Mechanical vibration signals contain recurring oscillatory components associated with rotor rotation and blade passage frequency. Detecting these periodic structures is important for understanding normal operating conditions and identifying deviations from expected patterns. Similarity analysis between signal segments may reveal repeated vibration signatures. The problem is to detect and interpret periodicity and recurring vibration patterns in the recorded signal using correlation methods.

Day 7 – Methodological

On this day, you will define and justify your methodological approach for analyzing both recurring structures and localized signal segments in your data. Based on your problem statement, explain why auto-correlation is appropriate for identifying global recurring patterns in your signal and why cross-correlation is appropriate for detecting a specific segment within the larger signal.

Describe which structural characteristics of your signal support the use of auto-correlation (e.g., periodic behavior, repeating cycles, long-range similarity) and which characteristics justify the use of cross-correlation for segment detection. Clearly distinguish between these two analytical objectives and explain how each method addresses a different aspect of your use case.

Outline the parameters you intend to use for both analyses, such as lag range, normalization strategy, window selection, or criteria for selecting the signal segment. Justify each methodological choice in relation to your signal properties and your defined objectives.

The focus of this day is on structured methodological reasoning and planning. Both auto-correlation and cross-correlation must be addressed. No computations or visualizations should be included.

Day 8 – Technical Implementation

On this day, you will implement the correlation analyses defined in your methodological planning (Day 7). Compute the auto-correlation of your selected signal to identify recurring structures or periodic patterns and visualize the auto-correlogram appropriately. In addition, extract the signal segment defined

in your plan and apply cross-correlation to detect its position within the full signal. Clearly indicate how the detected position is identified from the correlation output.

Use the parameter configuration defined in on Day 7 as your starting point. If adjustments are required during implementation, you may refine parameters using a trial-and-error approach to ensure correct and stable computation. At this stage, systematic parameter space exploration is not required.

The objective of this day is a technically correct, structured, and traceable implementation of both auto-correlation and cross-correlation. Quantitative parameter comparison, robustness testing, and structured reflection will be addressed in the following days.

Day 9 – Evaluation

On this day, you will systematically evaluate the performance of your correlation analyses. Define 1-2 quantitative evaluation approaches (e.g., numerical metrics and/or quantitatively interpretable visualizations) for assessing both auto-correlation and cross-correlation results. Clearly justify why the selected approach is appropriate for evaluating periodicity detection or segment localization in your specific signal and use case, and specify what aspect of performance it captures (e.g., peak detection accuracy, lag precision, robustness to distortion).

Using the defined evaluation approach, explore the relevant parameter space of your methods in a structured and traceable manner. Systematically vary 2-4 key parameters (e.g., lag range, normalization strategy, segment length) and quantify how performance changes across configurations relative to a clearly defined baseline. The same evaluation approach must be applied consistently across all configurations, and all reported quantities must include appropriate units or normalization.

For cross-correlation, conduct a robustness analysis by introducing 1-2 perturbations that are realistic and directly relevant to your defined use case (e.g., additive noise, amplitude variation, temporal shift). Recompute the cross-correlation and quantify how detection performance changes under these modified conditions using the same predefined evaluation approach.

The objective of this day is systematic, traceable, and quantitative evaluation of parameter choices and robustness. No new simulations beyond the defined perturbations, theoretical derivations, or interpretative discussion should be included at this stage.

Day 10 – Analytical & Communication

On this day, you will observe, interpret, and discuss your findings in direct relation to your original problem statement and defined use case. Based on your evaluation results, analyze how effectively auto-correlation revealed recurring structures and how reliably cross-correlation detected the selected signal segment.

Structure your analysis into observations, interpretation, and discussion. Examine the practical applicability of your results: Under which conditions does cross-correlation provide stable and reliable detection in your domain? How do the tested perturbations affect detection performance, and what does this imply for real-world use?

Critically analyze the relationship between your methodological design (plan), your implementation, and your evaluated parameter configuration. If adjustments were necessary, explain why and what this reveals about applying correlation methods in practice.

Present your conclusions concisely and in a technically precise manner (approximately 180-220 words). Your discussion must clearly connect quantitative results, parameter choices, robustness findings, and domain relevance. No new computations should be introduced.

1.3 Convolution and Deconvolution

Learning Objectives

- Conceptual: Apply convolution as a linear system operation and apply deconvolution as an inverse problem in a domain-specific context.
- Methodological: Select appropriate convolution kernels and a suitable deconvolution approach aligned with signal characteristics and performance requirements.
- Technical: Implement one-dimensional convolution manually and apply deconvolution in a structured and traceable manner.
- Analytical: Quantitatively compare original, filtered, and reconstructed signals using appropriate performance metrics.
- Reflective: Evaluate trade-offs between smoothing, stability, noise sensitivity, and reconstruction fidelity in practical applications.

Day 11 – Data & Domain

On this day, you will define the problem statement and use case in which one-dimensional convolution is an appropriate approach. Select a country-related signal and describe its origin, physical meaning, and characteristics. Clearly explain what issue you aim to address (e.g., smoothing, noise reduction, feature enhancement) and why convolution in the spatial domain is suitable for this objective.

Describe the expected effect of filtering on your signal and explain what improvement or transformation you seek to achieve. In addition, consider the role of deconvolution as the inverse operation of convolution. Explain how reconstructing or restoring a signal from its filtered or blurred version could be relevant in your domain (e.g., recovering attenuated features, reversing smoothing effects, restoring signal structure).

The focus of this day is on domain framing, problem formulation, and justification of convolution and deconvolution as meaningful analytical tools in your specific use case. No implementation or quantitative analysis should be included.

Start a new notebook for this block.

Reference examples: The following use cases are provided for orientation. You may select one of them and adapt it to your chosen country, or define a suitable alternative within your selected domain.

Ocean Wave Monitoring – Signal Smoothing and Feature Restoration

Use Case: At a coastal holiday destination, wave height is continuously measured to characterize sea conditions. The recorded signal reflects periodic swell motion combined with short-term wind-driven fluctuations and measurement noise.

Problem Statement: Wave height measurements often contain high-frequency noise and small-scale fluctuations that obscure dominant swell patterns. Convolution with a suitable smoothing kernel can attenuate short-term variations and highlight the underlying wave structure. However, smoothing also modifies peak amplitudes and may blur sharp transitions associated with sudden wave groups. In this context, deconvolution becomes relevant as a means of restoring attenuated features or approximating the original signal structure after filtering. The problem is to justify convolution as a tool for enhancing interpretability while acknowledging the limitations and potential instability of reconstructing the original wave dynamics.

Speech Signal – Noise Reduction and Signal Restoration

Use Case: A speech recording obtained in a noisy environment is processed to improve intelligibility. The signal contains broadband speech components superimposed with environmental noise.

Problem Statement: Speech signals recorded in real-world conditions are often degraded by additive noise or mild reverberation. Convolution with a smoothing or low-pass kernel can reduce high-frequency noise and enhance perceptual clarity. However, filtering may also attenuate consonant components and reduce spectral sharpness, altering phonetic structure. Deconvolution is relevant when attempting to reverse known filtering or reverberation effects in order to restore speech characteristics. The problem is to justify convolution as a method for controlled signal modification and to assess the conceptual feasibility of reconstructing the original signal from its filtered form.

Fitness Tracker – Motion Artifact Reduction and Recovery

Use Case: During a hiking holiday, a wearable fitness tracker records optical pulse (PPG) and acceleration signals. These measurements contain physiological rhythms as well as motion-induced disturbances.

Problem Statement: Physiological signals from wearable devices are frequently contaminated by motion artifacts and high-frequency disturbances. Convolution with an appropriate kernel can smooth the signal and suppress unwanted fluctuations, thereby improving the visibility of periodic components such as heartbeat or cadence. However, filtering may distort peak shapes or reduce temporal precision of physiological events. In situations where filtering parameters are known, deconvolution may serve as an inverse operation to approximate the original signal dynamics. The problem is to frame convolution as a tool for noise reduction while recognizing the challenges of accurately restoring physiological signal structure.

Wind Turbine Vibration Monitoring – Feature Enhancement and Reconstruction

Use Case: Vibration sensors installed in a wind turbine record mechanical oscillations associated with rotational motion and structural resonance. The signal may contain broadband noise and transient disturbances.

Problem Statement: Mechanical vibration signals often contain noise components that obscure characteristic frequency signatures of rotational dynamics or emerging faults. Convolution with a smoothing or band-limiting kernel can enhance dominant oscillatory behavior and suppress irrelevant fluctuations. However, filtering may attenuate transient fault signatures or alter amplitude relationships between spectral components. If the filtering process is well characterized, deconvolution may be considered to recover or approximate original vibration features. The problem is to justify convolution as a linear system operation for signal enhancement and to reflect on the feasibility and limitations of reversing such modifications.

Day 12 – Methodological

On this day, you will define and justify your methodological choices for convolution and deconvolution in direct relation to your problem statement and use case.

Select one or two suitable convolution kernels (e.g., moving average, Gaussian smoothing, or another justified filter) and explain why they are appropriate for your signal characteristics and objective. Clearly justify your choice of kernel type, kernel size, and any relevant parameters. Your reasoning should focus on how these choices influence filtering behavior in the context of your specific use case (e.g., smoothing strength, feature preservation, attenuation of noise).

In addition, research different approaches to deconvolution that are applicable to one-dimensional signals (e.g., inverse filtering, regularized inverse filtering, Wiener filtering, or another suitable method). Based on this comparison, select one deconvolution approach and justify why it is methodologically appropriate for your signal characteristics and anticipated challenges. Your justification should explicitly address expected reconstruction quality in your use case. How accurate does the reconstruction need to be in order to be practically useful? Discuss considerations such as stability, sensitivity to noise, reconstruction fidelity, and parameter requirements, and explain how these aspects influence your methodological choice.

The focus of this day is on justified method and parameter selection aligned with your use case and expected performance requirements. No algorithmic step descriptions, implementations, or experimental results should be included.

Day 13 – Technical Implementation

On this day, you will implement your convolution and deconvolution methods as defined in your methodological plan (Day 12).

First, implement one-dimensional convolution manually in the spatial domain. The primary convolution must be your own implementation; library functions may only be used for verification. Your implementation should clearly reflect the essential elements of convolution, including kernel application, sliding window operation, summation of products, a defined boundary handling strategy, or other relevant parameters. Apply your convolution to the selected signal and compare the result to an appropriate library implementation to verify correctness. Briefly observe whether the outputs match and explain any discrepancies.

Next, apply your selected deconvolution to the filtered signal to obtain a reconstructed version. For this part, appropriate library implementations may be used; however, ensure that parameter choices are transparently reasoned.

The objective of this day is technically correct, transparent, and well-structured implementation of both convolution and deconvolution. Systematic performance comparison and interpretation will be conducted on the following days.

Day 14 – Evaluation

On this day, you will systematically evaluate the effects of convolution and deconvolution in relation to the performance expectations defined previously in your methodological planning.

Begin by defining 1–2 quantitative evaluation approaches (e.g., numerical metrics and/or quantitatively interpretable visualizations) to assess filtering performance and reconstruction quality. Clearly justify why the selected approach is appropriate for your signal characteristics and use case, and specify which aspect of performance it captures (e.g., noise reduction, amplitude preservation, reconstruction fidelity, stability).

Using the defined evaluation approach, compare the original signal, the filtered signal, and the reconstructed signal in a structured and traceable manner relative to a clearly defined baseline configuration. Quantify smoothing effects, distortion, or information loss introduced by convolution and assess how effectively deconvolution restores relevant signal characteristics. The same evaluation approach must be applied consistently across all configurations, and all reported quantities must include appropriate units or normalization.

Systematically vary 2-4 key parameters (e.g., kernel size, smoothing strength, regularization parameter) and quantify how these choices influence filtering and reconstruction performance relative to the baseline. The objective is to analyze trade-offs between stability, noise sensitivity, and reconstruction fidelity using structured, quantitative comparison.

The focus of this day is structured, quantitative evaluation and justified comparison. No new theoretical derivations or interpretative discussion should be introduced at this stage. No extended interpretation or reflective discussion should be included at this stage.

Day 15 – Analytical & Communication, Final Reflection & Revisions

On this day, you will first observe, interpret, and discuss your convolution and deconvolution results in direct relation to your original problem statement and defined use case. Based on your quantitative evaluation, analyze whether convolution achieved the intended effect (e.g., smoothing, noise reduction, feature enhancement) and whether the selected deconvolution method provided meaningful reconstruction.

Structure your analysis into observations, interpretation, and discussion. Critically assess whether the achieved reconstruction quality meets the expectations defined in your methodological design (plan). Evaluate the practical applicability of your approach: Under which conditions is convolution beneficial in your domain? When does deconvolution meaningfully improve signal interpretation, and where do limitations such as instability, noise amplification, or information loss appear? If differences occurred between your methodological plan, implementation, and evaluated parameter configuration, explain their causes and implications.

Present this discussion concisely and in a technically precise manner (approximately 180-220 words). All arguments must be grounded in your quantitative evaluation results. No new computations should be introduced.

In addition, write a final reflection on MC1 (approximately 150 words). Reflect on what worked well, what you would approach differently, what aspects were particularly insightful, and what you learned about applying theoretical signal processing methods in practice.

Submit revisions, if applicable. Revisions are only considered if clearly highlighted in color. Code does not need to be colored; however, include a clearly marked text note in color before and after any changed code sections.